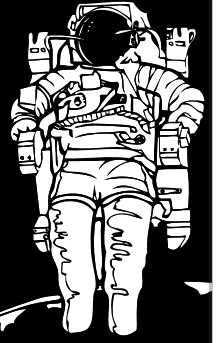


Presentation

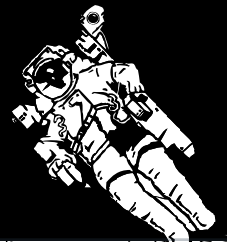
MOON EVOLUTION ODYSSEY





OBJECTIVES

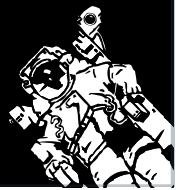
- Why are we doing this project?
- What can we expect to learn?
- Why are we analyzing moons?





CLASSIFICATION OF CELESTIAL OBJECTS

How can we tell whether the small dot visible in the night sky is a planet, a star, or a moon?



BASIC DIFFERENCE

Feature	Moons	Planets	Stars
Orbit	Orbit planets or dwarf planets	Orbit stars	Exist independently or in systems
Light Production	Reflect sunlight	Reflect sunlight	Produce their own light through fusion
Energy Source	None	None	Fusion of hydrogen into helium
Size and Mass	Small (relative to planets)	Intermediate (larger than moons)	Massive (larger than planets)
Examples	Earth's Moon, Titan	Earth, Jupiter	Sun, Betelgeuse

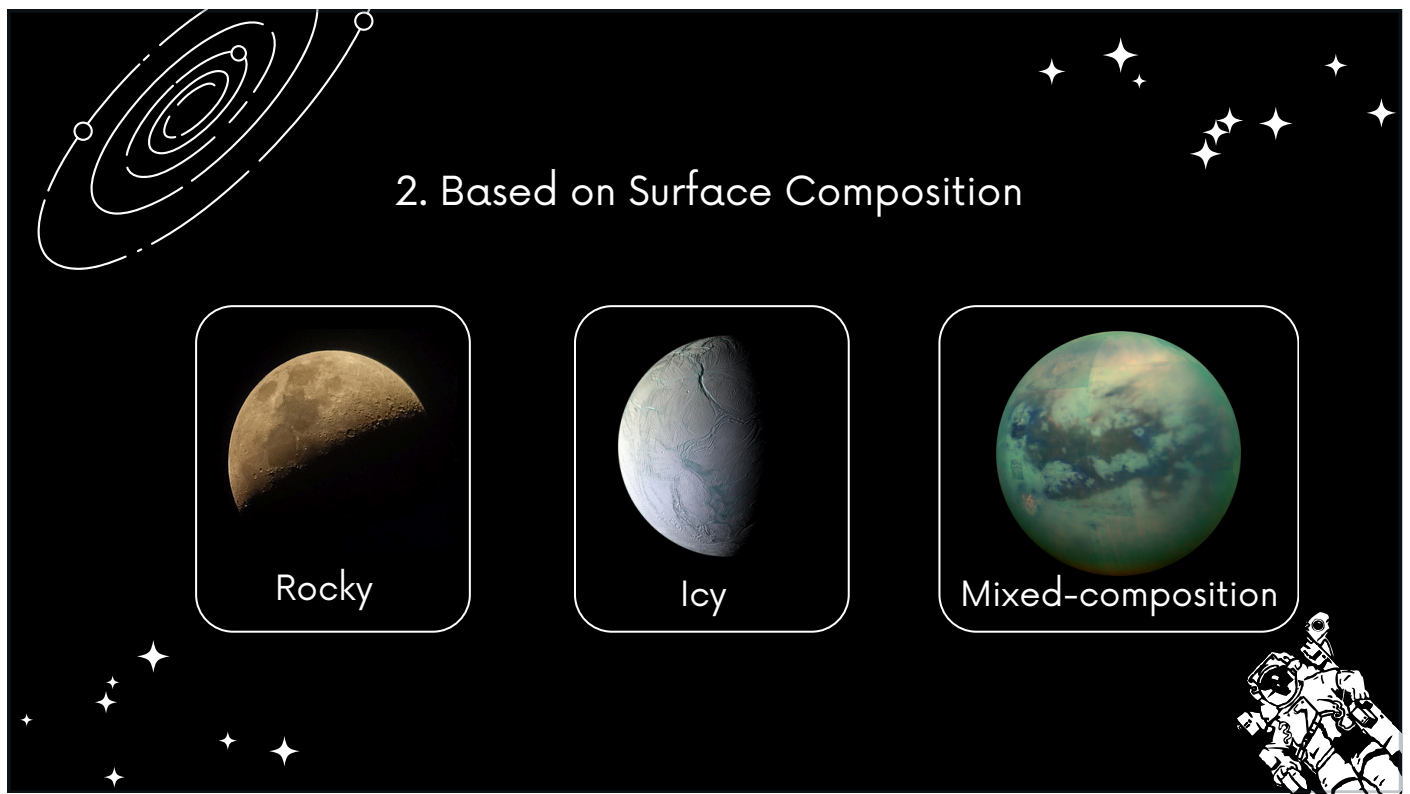


SUB CLASSIFICATION OF MOONS

1. Based on Physical Characteristics



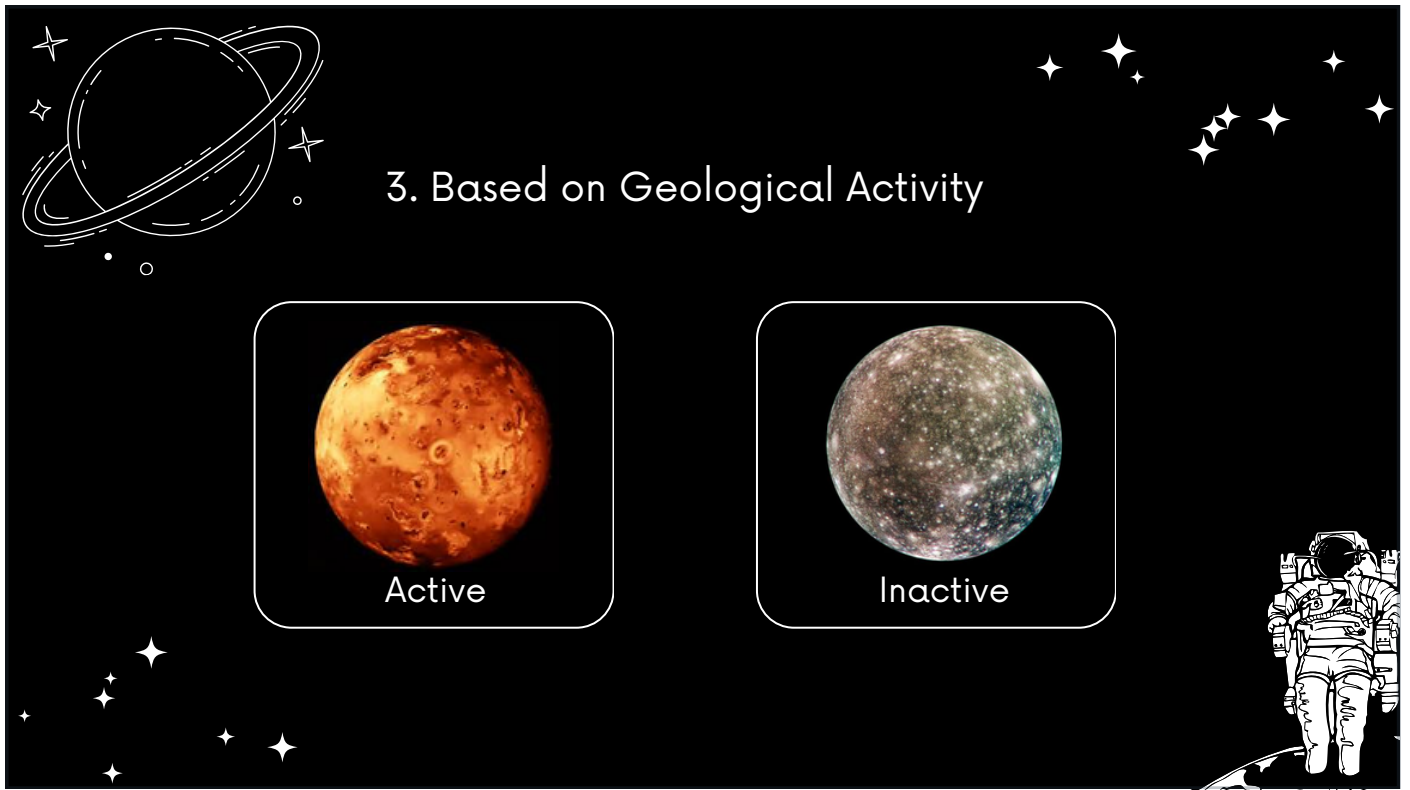
regular- Europa
irregular- phoebe
shepherd - pandora
trojan - calypso



Rocky- earth's moon

icy- Enceladus

mixed- titan



active- io

inactive - callisto

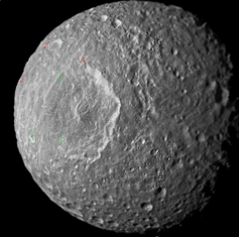
4. Based on Relationship with Parent Planet



Major



Minor

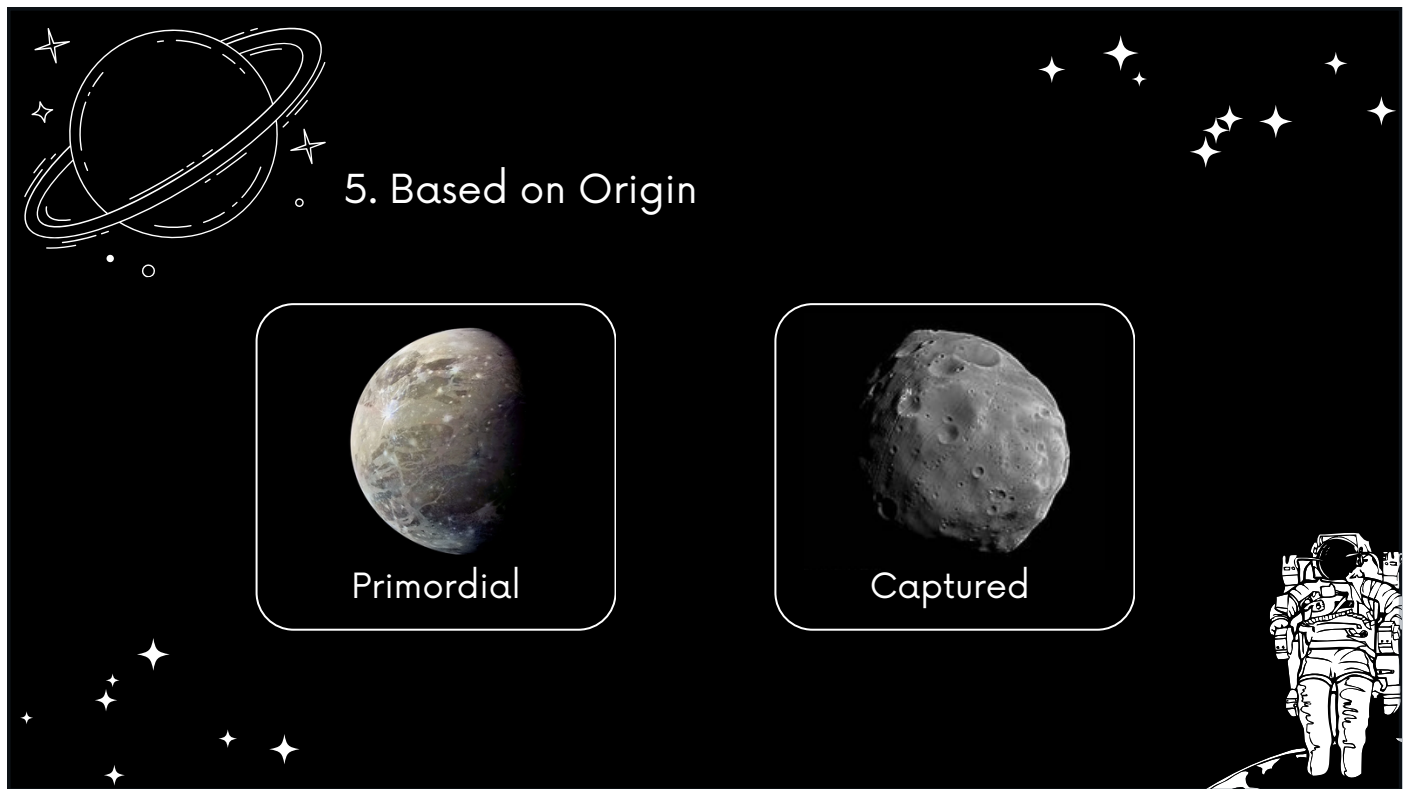


Moonlets

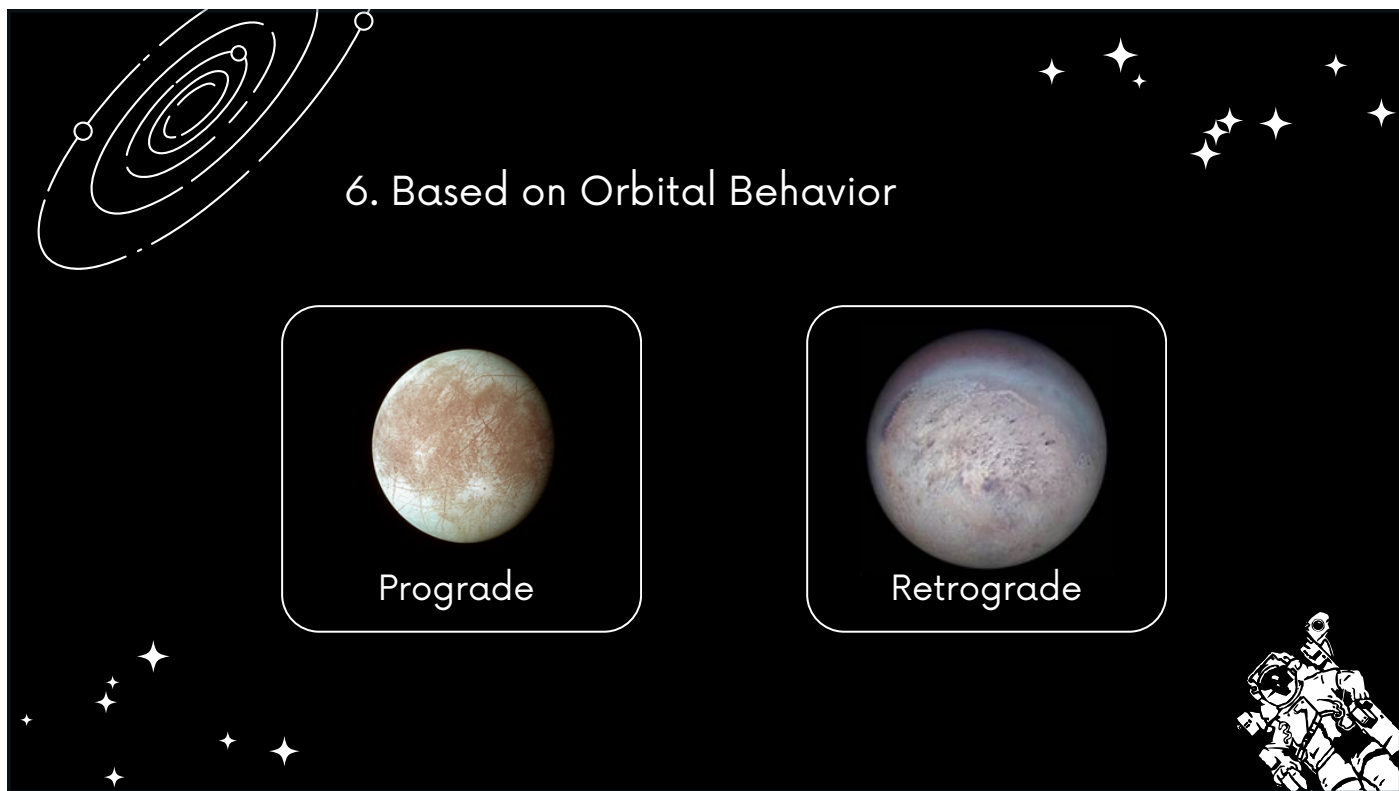
major- earth's moon

minor - Deimos

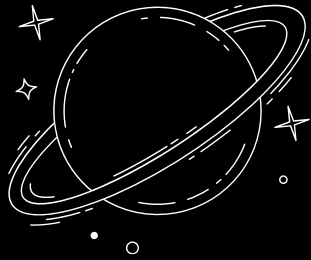
moonlets - Daphnis



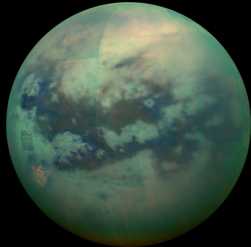
primordial- Ganymede
captured- phobos



prograde- Europa
retrograde - triton



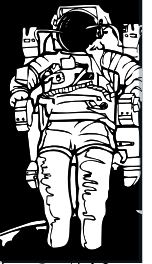
7. Based on Atmosphere



With atmosphere



Without atmosphere



with atmosphere - titan

without atmosphere - Callisto

The image shows a large, grey, cratered celestial body, likely a moon or asteroid, in the foreground on the left. The surface is covered in numerous craters of various sizes. In the background, a bright sun is visible in the upper right corner, casting a soft glow. The entire scene is set against a dark, starry space background.

MOONS OF OUR SOLAR SYSTEMS

Moon of Earth

"the Moon"

- The 1st moon to be discovered
- The 5th largest moon in the Solar system
 - Radius: 1 737.5 kilometers (3.7x smaller than Earth)
- The only place where a human being has been beyond Earth
- Surface has numerous craters and light areas (highlands) and dark areas (maria) that represent rocks of different composition and ages and are also old basins of lava
- Temperature varies between 127 °c and -173 °c



Moons of Mars

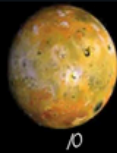
- Mars has two moons; Phobos and Deimos
- Both were discovered in 1877
- Smallest moons
- Phobos is a bit larger than Deimos and it orbits Mars three times a day
- Deimos takes about 30 h to orbit
- Both have a lumpy structure and are heavily-cratered and covered in dust and loose rocks



Moons of Jupiter



- Jupiter has 79 moons
 - 53 named and 26 that are waiting their official names
- The 4 biggest moons of Jupiter and the first objects found to orbit another planet are called the **Galilean moons** because they were found by Galileo Galilei in 1610. All of them are bright enough to be viewed from Earth without a telescope.
 1. **Ganymede** is the biggest moon in the whole Solar system (bigger than Mercury). It's the only moon known to have internally generated magnetic field. It has an icy surface .
 2. **Callisto** is the 2nd largest of the Galilean moons and the 3rd largest moon in the Solar System. It's is similar to our Moon because the surface is also heavily cratered.
 3. **Io** is the 3rd biggest Jupiter's moon and the 4th largest moon of the Solar system. It has the most volcano activity, the highest density and the least amount of water of all the moons in our Solar system.
 4. **Europa** is the smallest moon of the Galilean moons and 6th largest in the Solar system. Its surface is extremely flat and it contains mainly water ice. Scientists thinks that Europa is the best place in the solar system to look for alien life.



Moons of Saturn

- Saturn has 53 confirmed moons and nine waiting confirmation.
1. Titan is Saturn's biggest moon and the only known world other than Earth where any sort of liquid collects on its surface. Its surface is so cold which means the moon's surface is water ice instead of rock. This means methane and ethane exist on **Titan**.
 2. Rhea is the second largest moon. It has a typical cratered surface. Its surface can be divided into two geologically different areas because half of its surface is based on crater density and the other half is icy.
 3. The third biggest moon **Iapetus** is not perfectly round it's a bit flatten. It also has one bright icy side and one very dark side.



Moons of Uranus

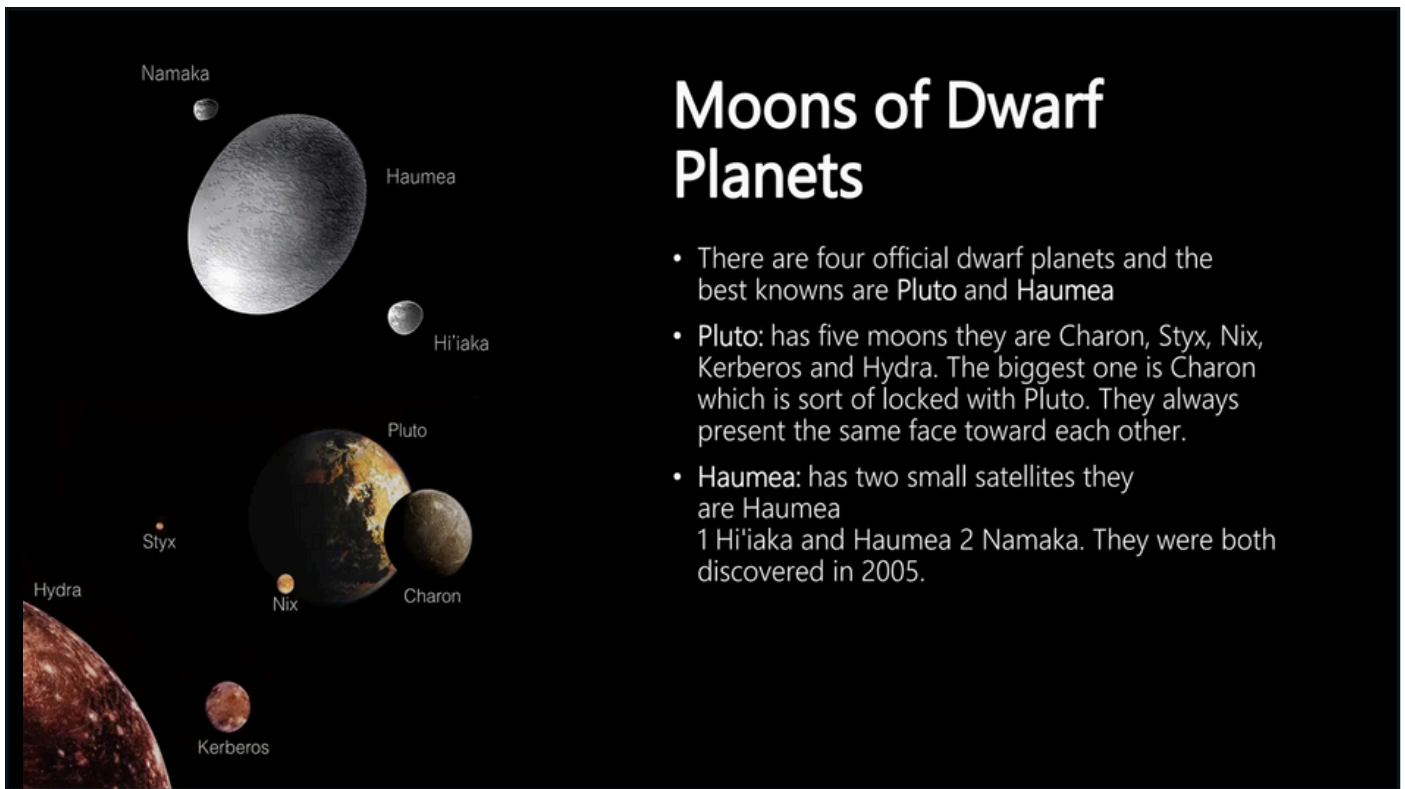


- Uranus has 27 natural satellites. The five main moons are Miranda, Ariel, Umbriel, Titania and Oberon.
1. The largest one Titania's surface is dark and lightly red in color and its covered with numerous impact craters.
 2. The outermost but the second largest Oberon has a red surface. But its fresh impact deposits are light blue. Oberon's surface is the most cratered of all the Uranian moons.
 3. Umbriel the 3rd largest moon of Uranus was discovered the same time as Ariel. Umbriel's surface is the darkest and reflects half as much light as Ariel and its light is blue.
 4. Fourth largest moon is Ariel. It's the most reflective Uranian moon. Its surface can be divided into three terrain types: cratered terrain, ridged terrain and plains.
 5. Miranda the smallest and innermost Uranus's satellite. It's surface is mostly water ice. It's the least spherical Uranian moon.

Moons of Neptune

- Neptune has 14 known moons.
1. Triton is the largest and the first Neptunian moon to be discovered. Almost half of Triton's surface has ridges, icy plains, troughs, furrows and few craters. The other half is covered with frozen nitrogen with water ice and dry ice.
 2. Proteus has a really dark surface and it was so close to Neptune that it was hard to discover.
 3. Nereid's shape is unknown. It's surface is composed of a mixture of water ice and some spectrally neutral material.





Moons of Dwarf Planets

- There are four official dwarf planets and the best knowns are Pluto and Haumea
- Pluto: has five moons they are Charon, Styx, Nix, Kerberos and Hydra. The biggest one is Charon which is sort of locked with Pluto. They always present the same face toward each other.
- Haumea: has two small satellites they are Haumea 1 Hi'iaka and Haumea 2 Namaka. They were both discovered in 2005.

